

Assignment Template: Project Assignment 1

Team Name: Project Healthify

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1. Functional Requirements

- **FR-01:** Take telemetry data from things like an HR monitor or step-tracking function using some sort of wearable to track these functions
- **FR-02:** Wearable Device Connection (i.e., Apple Watch, Whoop, Ouro, etc.)
- **FR-03:** Alerts instilled to let the user know when things like their telemetric data (HR, HRV, hours slept, blood pressure, calories burnt, changes in activity level) show abnormalities either in general metrics or from calculating user parameters
- **FR-04:** EMS Autodetection (HRV -> AFib, sudden fall, slow heart rate, etc)
- **FR-05:** Allow the user to enter their info(gender, age, weight) to calculate their health metrics
- **FR-06:** Allow the user to register a new account using an email/password, making sure credentials are entered through a secure database
- **FR-07:** Data Storage and Retrieval for the user (Cloud)
- **FR-08:** Reporting and Analytics for the user (Daily or weekly recap)
- **FR-09 [Added after GenAI]:** An emulator for wearable telemetry in case of integration failures or difficulty obtaining a partnership agreement with a wearable company.

2. Non-Functional Requirements

- **NFR-01:** Fast, reliable user interface for a seamless user experience (<1 UI response times)
- **NFR-02:** Allow the user to choose a goal(like weight loss, increased step count, less caloric intake, etc.) and offer the user reminders or measurable improvement to the user.
- **NFR-03:** Personal health and account information is encrypted.
- **NFR-04:** Scalability; The app should be able to handle multiple users simultaneously.

- **NFR-05:** Cache user data to measure changes in telemetry day-to-day, be it caloric intake, steps, or HR

3. User Stories with Acceptance Scenarios

US-01 [Modified after GenAI]: Device Connection

As a Registered User, I want to pair my Health Tracking Device so that my telemetric data automatically synchronizes my health profile.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given I am on the settings page and select "Connect Device"

When I am redirected to my wearable provider's login page and authorize access to my account

Then the system stores my authorization token, displays a "Device Connected" status, and syncs my available telemetry data

Scenario 2: Negative/Edge Case

Given my wearable account is connected

When the system's scheduled sync makes a request to the provider's API and the request fails (expired authorization, provider outage, or rate limit reached)

Then the system logs the failure, displays a "Reconnect Required" or "Sync Failed" indicator, and retries on a backoff schedule rather than repeatedly hammering the API

US-02: Sign-in screen

As a Unregistered User, I want to create a secure account using my own email/password so that my data could be securely stored in some sort of cloud database.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given I am on the system registration page

When enter a valid email/password that meet a secure requirements(one capital letter, one special character, one number)

Then after creating an account be redirected to a dashboard

Scenario 2: Negative/Edge Case

Given I am on the system registration page

When I try to register with an email/password that's already in the database

Then an error message should pop up asking to create an account using different login credentials

US-03: Dashboard

As a User, I want a main page with relevant information so that I can access features and my information.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given I am a registered user on the login page.

When I login with my valid credentials.

Then I want to see the dashboard displaying key information and functionalities.

Scenario 2: Negative/Edge Case

Given I am a registered user navigating directly to the main dashboard.

When my active session token expires due to inactivity.

Then I want the system blocking access to the dashboard, redirecting me to the main login screen, and displaying a message like "Session Expired, please re-login."

US-04: Health Abnormality Alerts

As a User, I want to receive immediate alerts when my wearable metrics show sudden abnormalities so that I can take prompt medical or preventative action.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given the user has active telemetry monitoring enabled on their wearable device

When the wearable detects a resting heart rate exceeding 130 BPM for over 5 minutes (for example)

Then the system sends an urgent push notification alerting the user to an abnormal heart rate spike

Scenario 2: Negative/Edge Case

Given the user's wearable loses connection while transferring telemetry data

When an abnormal metric spike occurs during the disconnection period

Then the system logs the data gap upon reconnection and prompts the user to check their device placement without triggering a false alarm

US-05: User Profile Parameters

As a New User, I want to enter my personal health information (gender, age, weight) so that the app can accurately calculate my baseline metrics and health targets.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given the user is on the Profile Setup screen

When the user enters valid personal details (Age: 29, Gender: Male, Weight: 175 lb, etc.) and clicks "Save Profile"

Then the system saves the attributes to the secure database and updates personalized health baseline metrics.

Scenario 2: Negative/Edge Case

Given the user is on the Profile Setup screen

When the user leaves the weight field blank or enters an invalid value (e.g., "-10") and clicks "Save Profile"

Then the system prevents form submission and displays an inline validation error asking for a valid number.

US-06: Goal Setting & Reminders

As a User, I want to define specific fitness goals and set up progress reminders so that I can track measurable improvements over time.

Acceptance Scenarios (Gherkin):

Scenario 1: Positive Flow

Given the user is on the Goal Management screen

When the user selects "Daily Step Count", inputs a target of (10,000 steps), and enables reminders

Then the system saves the target goal and schedules progress notifications when milestones are reached

Scenario 2: Negative/Edge Case

Given the user sets a daily step goal of 10,000 steps

When the day ends, and the user's step count remains below 5,000 steps

Then the system logs the goal as incomplete for the day and generates a supportive motivational prompt for the next morning

4. Brainstorming and GenAI Reflection

Step A: Team Brainstorming

- Decision Point 1: Implement only what is specified in the description of the project definition (no third-party integration for a food tracker or an exercise tracker).
- Decision Point 2: The app should be able to deal with multiple users whose data are encrypted.
- Decision Point 3: EMS autodetection vs. health “abnormality”. An “abnormality” is defined as a change in health metrics, whereas an emergency situation is triggered by high heart rate variability, low heart rate, fall detections (if a wearable device has an accelerometer). Additionally we agreed to simulate calls to emergency services.

Step B: GenAI-Assisted Brainstorming

- GenAI Suggestion 1: Merge **FR-01** and **FR-02** into one functional requirement which includes connection to and telemetry from a wearable technology. Be specific in which wearable technology we want to work with, as the Apple Watch requires an actual Apple Watch as well as a \$100 a month developer kit, and Whoop and Oura similarly require applications to their respective companies to gain developer access. Initially GenAI suggested Fitbit, though over the course of this assignment Fitbit has announced though will be deprecating their Web API in the next month. The best real path and integration technology at this point is either Google's Health API or Whoop. However the amount of churn in this sector underscores the need for some telemetry emulator.
- GenAI Suggestion 2: Medical emergency detection and EMS autodialing (**FR-04**) is an overscoped feature. Detecting a fall with accelerometers or making determinations about atrial fibrillations from heart rate variability (HRV) are difficult engineering problems which took companies like Apple and Fitbit years to solve. Additionally SMS messaging would require a paid third-party API (e.g. Twilio) which we would like to avoid paying for. Replace

with simple notifications in-app to the user about abnormalities in baseline, highlighting that all in-app information should not be taken as real medical advice.

- GenAI Suggestion 3: Caching data as per **NFR-01** is likely unnecessary as we initially intended that specifically for a food logger functionality to reduce latency times between API calls for common food items, and likely isn't necessary for other user information. Either keep the food logger feature, as the description for the project outlined in the course materials is not the SSOT for the design of our project, or replace it with a non-functional requirement which spells out how to handle failures with third-party data from wearables or any other source. Potentially include an emulator for that data as a backup which can be used in place of live data.

Step C: Refined Requirements & Stories

- **FR-09 [Added after GenAI]:** An emulator for wearable telemetry in case of integration failures or difficulty obtaining a partnership agreement with a wearable company.
- **US-01 [Modified after GenAI]:** Our wearable integration depends on which device we pick. We've already planned emulators as a fallback in case we can't use Whoop's or Fitbit's (now Google Health's) REST APIs. If we do use one of those, we're not connecting via Bluetooth — we authenticate with OAuth 2.0 and pull data over the web instead. US-01 needs to reflect that.

US-01 [Modified after GenAI]: Device Connection

As a Registered User, I want to connect my wearable provider account so that my telemetry data automatically syncs to my health profile.

Scenario 1: Positive Flow

Given I am on the settings page and select "Connect Device"

When I am redirected to my wearable provider's login page and authorize access to my account

Then the system stores my authorization token, displays a "Device Connected" status, and syncs my available telemetry data

Scenario 2: Negative/Edge Case

Given my wearable account is connected

When the system's scheduled sync makes a request to the provider's API and the request fails (expired authorization, provider outage, or rate limit reached)

Then the system logs the failure, displays a "Reconnect Required" or "Sync Failed" indicator, and retries on a backoff schedule rather than repeatedly hammering the API

Step D: Reflection

GenAI improved coverage of requirements, user stories and scenarios. It performed this by identifying gaps, showing unnecessary complexity when the original requirements were too broad. The initial requirements included wearable connections, health abnormality alerts, cloud storage, and user goals. GenAI helped focus down some of those features into more manageable chunks or told to scale back on a particular feature. This allowed for better scope of the project and focus on more realistic features for the project's development.

The new perspectives from GenAI suggestions showed the importance of selecting a platform based on developer usability rather than choosing a popular device. Most popular devices have proprietary software that can make it very expensive to use. GenAI also pointed out that the caching requirement was based on an early idea using food tracking, that may or may not be currently part of the project. This allowed for more requirements to be continually reviewed as the project scope changes

The use of GenAI in requirement gathering should be used as a brainstorming and review tool rather than a replacement for the team's decisions. The GenAI can identify the possibilities that a team may overlook, however suggestions still need to be evaluated based on the team capabilities and skills.